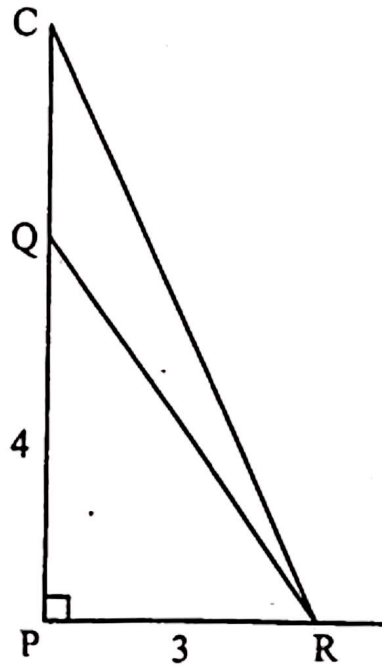




In the diagram $\hat{CPR} = 90^\circ$, $PR = 3\text{cm}$ and Q is the point on PC such that $PQ = 4\text{cm}$.

- (a) Calculate QR .
- (b) Write down the value of $\cos \hat{CQR}$.
- (c) Given also that $QC = \frac{1}{2}PQ$, calculate the area of $\triangle CQR$.

Answer

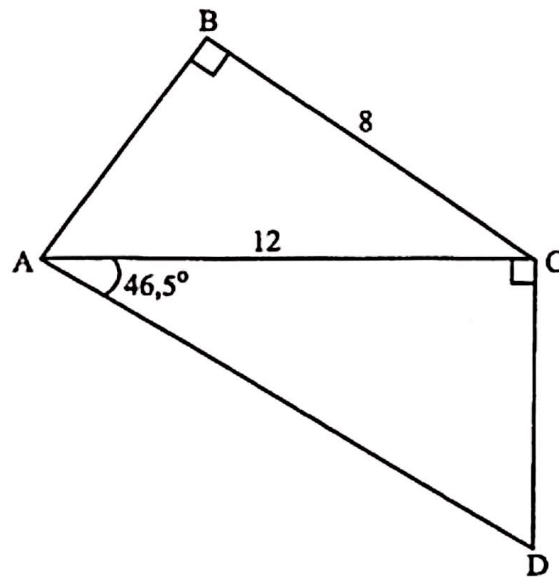
(a) $QR = \underline{\hspace{2cm}}\text{cm}$ [1]

(b) $\cos \hat{CQR} = \underline{\hspace{2cm}}$ [1]

(c) Area of $\triangle CQR = \underline{\hspace{2cm}}\text{cm}^2$ [1]

4

2 (a)



The diagram shows a quadrilateral ABCD with $BC = 8$ cm, $AC = 12$ cm
 $\angle CAD = 46,5^\circ$ and $\angle ABC = \angle ACD = 90^\circ$.

(i) Calculate AB,

Answer (i) $AB =$ _____ cm [2]

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2 (a) (ii) Calculate CD.

Answer (ii) CD = _____ cm [2]

2 (a) (iii) Calculate the area of quadrilateral ABCD.

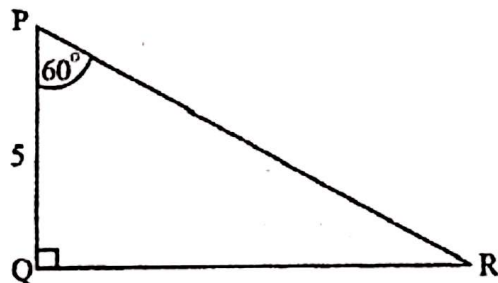
Answer 2 (a) (iii) _____ cm² [3]

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In the diagram $\angle PQR = 90^\circ$, $PQ = 5$ cm and $\angle QPR = 60^\circ$.

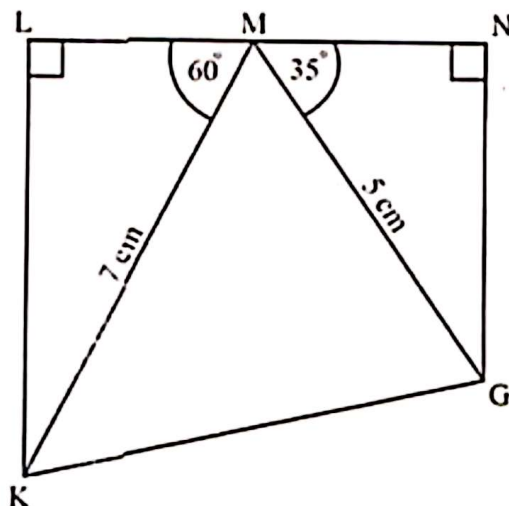
- (a) Calculate PR.

[Use as much of the information given below as is necessary]

[$\sin 60^\circ = 0,866$; $\cos 60^\circ = 0,500$; $\tan 60^\circ = 1,732$]

Answer: (a) $PR =$ _____ cm [2]

10



In the diagram $KLNG$ is a trapezium in which LMN is a straight line and $\angle KLM = \angle MNG = 90^\circ$. $MK = 7$ cm, $MG = 5$ cm, $\angle LMK = 60^\circ$ and $\angle MNG = 35^\circ$. Calculate the

- (a) length of NG .

Answer (a) [2]

- (b) size of $\angle KMG$.

Answer (b) [1]



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(c) area of triangle KMG,

Answer (c) [2]

(d) length of KG,

Answer (d) [4]

(e) size of $\hat{MKG}$.

Answer (e) [3]

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